

Saber for Automobile System Design and Simulation

Included in this month's EFY DVD is Saber simulation tool for automotive system design. It is a platform for modelling and simulating physical systems, enabling full-system virtual prototyping for applications in analogue/power electronics, electric power generation/conversion/distribution and mechatronics

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Decades of industry use and innovation have earned Saber a reputation as one of the most reliable solutions for design validation and optimisation in various domains like automotive, aerospace, industrial power and energy systems. Here we discuss the applications and solutions that Saber brings to design engineers' tables in automotive domain.

Simulation in automotive sector

The economics of automotive industry are becoming more challenging for car makers. Increased pressures for productivity and quality, and proliferation of complex electrical systems are just some of the issues that the industry needs to address. Quality is a significant challenge for automotive industry. The root causes of quality issues are often the designs that do not account for sources of variation, including component tolerance, environmental conditions, component ageing and so on. Statistical analysis using actual parts is economically prohibitive due to the size of systems involved. Compounding the quality challenge is the proliferation of electrical subsystems.

The electrification of vehicles has made them so complex that traditional design approaches don't achieve performance and design goals. This trend is driving an increase in projects requiring simulation and analysis. On top of these challenges, automo-

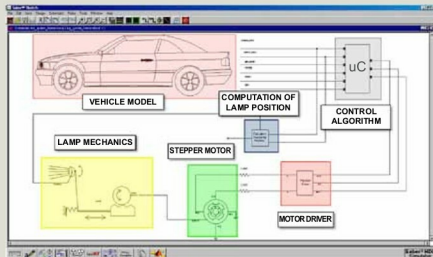


Fig. 1: Sketch of head lamp

tive industry has had to deal with economic pressure in recent years. Reductions in engineering budgets have led to an increase in workload that has to be managed by the same (or sometimes reduced) levels of staffing. Competition from global players has intensified, putting more pressure on development time scales as manufacturers strive to bring their products to market faster.

Robust design methodology

Successful automotive design requires a systematic method to meet rapidly changing and increasingly aggressive performance goals—not to mention compressed schedules. Saber is used to design and verify the interaction of multiple technologies, viz, electrical, mechanical, hydraulic, magnetic and software among

numerous others. Designers create virtual prototypes of the system, including the wire harness, to reduce the number of design iterations and hardware prototypes. Far more tests and design variations are possible in this environment than the regular hardware prototyping.

Through a robust design methodology, system performance is immunised against variance in parameters. These methods have proven to reduce design cycle time and maximise performance and reliability. All systems are subject to sources of variation including component tolerances, environmental stresses and ageing. Automotive system engineers want to reduce the effects of variability on system performance by designing systems that are less sensitive to these sources of variation. Saber helps engineers to apply robust